

Day 29 - First Hop Redundancy Protocols

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FHRP Information

A First Hop Redundancy Protocol (FHRP) is a networking protocol, which is designed to protect the *default gateway* used on a subnet by allowing two or more routers to provide backup for that address. If a failure does occur on the *active* router, the *backup* router will take over the address.

- **FHRP** works by assigning routers in the network a **Virtual IP (VIP)** address which will be set as the *default gateway* on all end-hosts, rather than the actual IP address of the routers in the network.
- **FHRP** also assigns the routers a unified **Virtual MAC Address** for the end-hosts to use in order to send frames to the routers. The **Virtual MAC** is decided by the **FHRP** protocol that's being used on the network.
- The routers in the **FHRP** negotiate and communicate with each other through *multicast Hello* messages, in order to decide their roles. (*Active & Standby / Master & Backup / etc.*)
- When a change in roles happens (ie. Router goes from *standby* to *active*), the new *active* router will broadcast **Gratuitous ARP** messages on the *ffff.ffff.ffff* MAC Address, which are basically **ARP Reply** messages that get sent without even being requested (through an **ARP Request**). This is used to update the switches' **MAC Address Tables**.

What do Gratuitous ARP messages do exactly?

- The problem that **Gratuitous ARP** fixes is that when a router goes from *standby* to *active*, the switches in the network won't automatically know that the current *active FHRP* router in the network has changed. Therefore, they will keep forwarding frames/packets to the former *active* router location.
- The solution to this was to make it so when a router gets assigned the *active* role in **FHRP**, it'll broadcast **Gratuitous ARP** messages that'll reach all the switches in the network; in order for them to get informed about the new *active* router's location and update their **MAC Address Tables** accordingly.

FHRP Routers and End-Hosts

FHRP Active and Standby Routers

If the currently *active* FHRP router goes down, the *standby* router will stop receiving *Hello* messages and assume that the *active* router has gone down.

- This is when it'll switch its **FHRP** role from *standby* to *active*.
- It'll also start sending out **Gratuitous ARP** to inform the switches in the network of the new *standby* router's location.

FHRPs are '*non-preemptive*'. Meaning that the current *active* router will not give up its role, even if the *former active* router gets reactivated. (Preemption can be enabled on different FHRP protocols, which would let the old router take back its role if it gets back up and running)

Process of End-host Communication with FHRP Routers

Usually, when an end-host wants to send a frame out of its subnet, it'll send it to its own *default gateway* which is the router interface's IP. However, when FHRP is set up, the *default gateway's* address is the Virtual IP (VIP) that was assigned onto the FHRP routers. Additionally, the end-host still needs a MAC address (which will be the Virtual MAC Address) so it could forward the frame to its *default gateway*.

- The end-host will send an **ARP Request** with the *source IP* being its own, and the *destination IP* being the **VIP**.
- Once the *active FHRP* router receives the request, it'll send back an **ARP Reply** containing the router's **Virtual MAC Address**.
- The end-host will then send traffic as normal to their *default gateway*. Frames that the end-host sends will have:
 - The *source IP* of the end-host
 - The *destination IP* which is the **VIP**
 - The *source MAC address* of the end-host
 - The *destination MAC address* which is the **Virtual MAC**

FHRP Protocols

FHRP Protocol Table Summary

FHRP Protocol	Terminology	Multicast IP	Virtual MAC	Proprietary?
HSRP	<i>Active/Standby</i>	V1: 224.0.0.2 V2: 224.0.0.102	V1: 0000.0c07.acXX V2: 0000.0c9f.fXXX	Yes
VRRP	<i>Master/Backup</i>	224.0.0.18	0000.5e00.01XX	No
GLBP	<i>AVG/AVF</i>	224.0.0.102	0007.b400.XXYY	Yes

1. HSRP (Hot Standby Router Protocol)

- Cisco proprietary
- **Active** and **Standby** routers are elected.
- In a situation with multiple subnets/VLANs, you can configure a different *active* router in each subnet/VLAN to load balance.
- Has two versions:
 - **Version 1**
 - Multicast IP: 224.0.0.2
 - Virtual MAC: 0000.0c07.acXX (*XX = HSRP group number*)
 - **Version 2** which adds *IPv6* support and increases the number of *groups* that can be configured
 - Multicast IP: 224.0.0.102
 - Virtual MAC: 0000.0c9f.fXXX (*XXX = HSRP group number*)

How the *active* role is assigned:

- The *active* router is determined in this order
 1. Highest **HSRP priority** (Default is **100**)
 2. Highest IP address

2. VRRP (Virtual Router Redundancy Protocol)

- Open standard
- **Master** and **Backup** routers are elected.
- In a situation with multiple subnets/VLANs, you can configure a different *master* router in each subnet/VLAN to load balance.
- Multicast IP: 224.0.0.18
- Virtual MAC: 0000.5e00.01XX (*XX = VRRP group number*)

3. GLBP (Gateway Load Balancing Protocol)

- Cisco proprietary
- Load balances among multiple routers within a single subnet
- An **AVG (Active Virtual Gateway)** is elected along with four **AVFs (Active Virtual Forwarder)**
 - The **AVG** itself can be an **AVF** as well.
 - Each **AVF** acts as the *default gateway* for a portion of hosts in the subnet.

- Multicast IP: 224.0.0.**102**
- Virtual MAC: 0007.b400.XXYY (XX = **GLBP AVG** group number, YY = **GLBP AVF** group number)

Extra: FRSP HSRP Configuration

HSRP is configured through the *standby* command.

- The **Virtual IP** can be assigned through this command:
 - *standby GROUP-NUMBER ip VIRTUAL-IP*
- The interface's **priority** can be set through this command: (default is 100, if both are same router with highest IP will be the active)
 - *standby GROUP-NUMBER priority PRIORITY-VALUE*
- Preemption can be enabled through this command:
 - *standby GROUP-NUMBER preempt*
- You can see the current **HSRP** configuration and settings on the router through this command:
 - *show standby*

Preemption in HSRP

Preempt causes the router to take the role of the **active** router, even if another router already has the role.

Basic HSRP Configuration

```
R1(config-if)# standby version 2
R1(config-if)# standby group-number ip virtual-ip
R1(config-if)# standby group-number priority priority
R1(config-if)# standby group-number preempt
```